



Lactate Dehydrogenase and Hemorrhagic Stroke During Extracorporeal Membrane Oxygenation for COVID-19

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Abstract

Purpose Hemorrhagic stroke (HS) is a devastating complication during extracorporeal membrane oxygenation (ECMO) but markers of risk stratification during COVID-19 are unknown. Lactate dehydrogenase (LDH) is a readily available biomarker of cell injury and permeability. We sought to determine whether an elevated LDH before ECMO placement is related to the occurrence of HS during ECMO for COVID-19.

Methods Adult patients with COVID-19 requiring ECMO between March 2020 and February 2022 were included. LDH values prior to ECMO placement were captured. Patients were categorized into high (> 750 U/L) or low (≤ 750 U/L) LDH groups. Multivariable regression modeling was used to determine the association between LDH and HS during ECMO.

Results There were 520 patients that underwent ECMO placement in 17 centers and 384 had an available LDH. Of whom, 122 (32%) had a high LDH. The overall incidence of HS was 10.9%, and patients with high LDH had a higher incidence of HS than those with low LDH level (17% vs 8%, $p=0.007$). At 100 days, the probability of a HS was 40% in the high LDH group and 23% in those with a low LDH, $p=0.002$. After adjustment for clinical covariates, high LDH remained associated with subsequent HS (aHR: 2.64, 95% CI 1.39–4.92). Findings were similar when restricting to patients supported by venovenous ECMO only.

Conclusion Elevated LDH prior to ECMO cannulation is associated with a higher incidence of HS during device support. LDH can risk stratify cases for impending cerebral bleeding during ECMO.

Keywords Extracorporeal Membrane Oxygenation · Lactate dehydrogenase · Hemorrhagic stroke · COVID-19

Abbreviations

ECMO Extracorporeal membrane oxygenation
ARDS Acute respiratory distress syndrome

HS Hemorrhagic stroke
LDH Lactate dehydrogenase

Introduction

Extracorporeal membrane oxygenation (ECMO) has had a critical role during refractory respiratory or cardiopulmonary failure from COVID-19 [1–4]. The majority of cases are indicated for acute respiratory distress syndrome (ARDS)-related respiratory failure with venovenous extracorporeal membrane oxygenation (VV-ECMO). Despite advances in medical treatment for COVID-19, mortality for patients requiring ECMO remains elevated, ranging from 30 to 55% [2, 5–8]. To improve outcomes, it is essential to identify baseline factors and biomarkers that may assist in risk stratification. Among factors associated with mortality, hemorrhagic stroke (HS) represents the single most important complication associated with death [3, 7, 9–18]. Moreover,

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small studies have indicated that the incidence of HS is elevated during ECMO for COVID-19 [19]. Early identification of patients at risk for HS could serve as a potential measure to ameliorate the incidence of this devastating complication and improve morbidity and mortality.

Lactate dehydrogenase (LDH) is a readily available biomarker of global tissue and cellular injury and is elevated during severe COVID-19 infection, associated with worse outcomes among these patients [20–22]. LDH has been used as a prognostication marker for neurological complications in non-ECMO settings, as it has been positively linked with increased endothelial damage and vascular permeability [21, 23–28]. However, its role in risk stratification for HS in COVID-19 cases treated with ECMO remains uncertain. Using a multicenter cohort of patients, we sought to determine whether an elevated LDH before cannulation is related to the occurrence of HS during ECMO support for COVID-19 infection.

Methods and Materials

Patient Selection

All adult patients undergoing ECMO support for respiratory and/or cardiac support with COVID-19 infection between March 2020 and February 2022 from participating centers

were included in this retrospective cohort analysis. Patients with no available data on LDH were excluded (Fig. 1). Data source was a voluntary registry populated by 17 participating centers located in different regions in the United States. Data entry, training and validation procedures of this registry have been published [29]. This study was approved by the institutional review board at montefiore medical center.

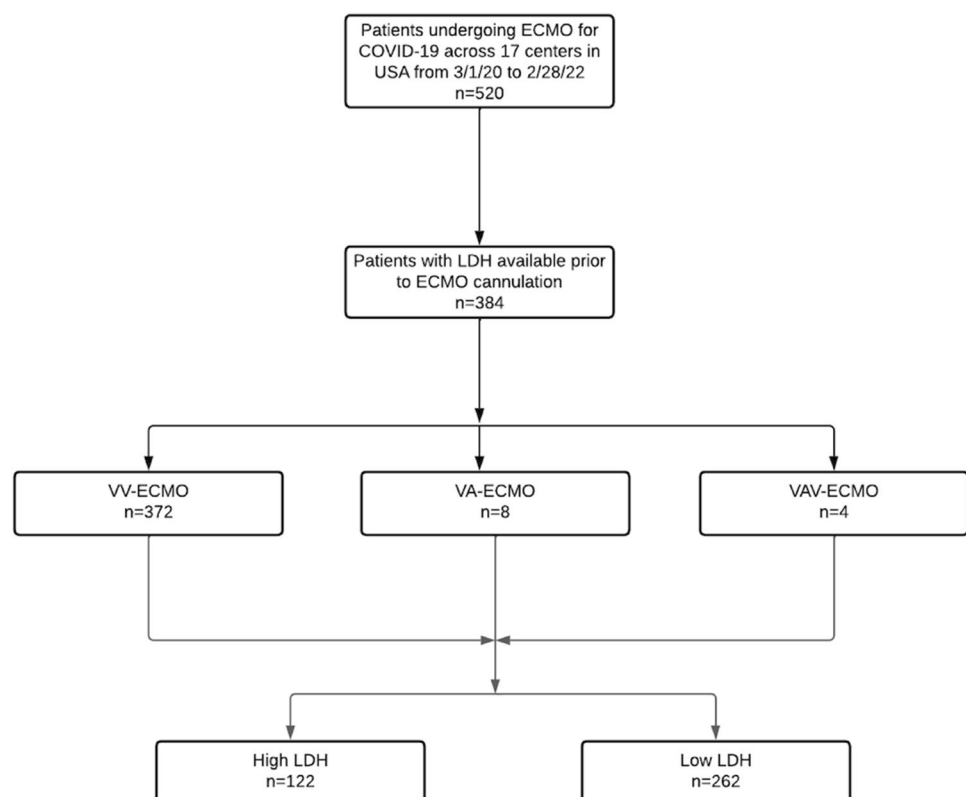
Exposure and Outcomes

The exposure of interest was serum LDH prior to ECMO cannulation. The study cohort was categorized into patients with high (> 750 U/L) or low (≤ 750 U/L) LDH. This threshold was derived by applying receiver operating characteristics curve analysis to identify a cut-point with optimal sensitivity and specificity for HS (sensitivity 0.61, specificity 0.60). The primary outcome was occurrence of HS, evidenced via brain imaging using head computed tomography as clinically indicated based on clinical judgement by the treated physician..

Statistical Analysis

Categorical variables are expressed as counts and percentages and were evaluated using the chi-squared test. Continuous variables are presented as means and standard deviations (SD) or medians and interquartile ranges (IQR)

Fig. 1 STROBE diagram showing formation of the study groups



as appropriate and compared between groups using Student's *T*-test or Mann–Whitney test according to the distribution of the variable. Kaplan Meier analyses was used to estimate and compare the probability of HS during ECMO between patients with a high or low LDH. Patients were censored at 100 days after ECMO initiation. Censoring was done for ischemic stroke, weaning off ECMO, device upgrade, or death. Multivariable Cox proportional hazard models were used to determine the risk of HS for patients with a high LDH in comparison those with a low LDH. Adjustment was performed for covariates present in > 90% of patients, previously known to be associated with HS and those with a univariable associated with LDH groups ($P < 0.2$). A subgroup analysis was performed for patients cannulated for venovenous ECMO. Adjusted hazard ratios (aHR) and 95% confidence intervals (CI) are reported. All statistical tests were two-sided and $p < 0.05$ was considered for statistical significance. All analyses were performed using Stata 14.0 (StataCorp LLC, Texas).

Results

Patient Characteristics

There were 520 patients that underwent ECMO placement in the study period, of which 384 had an available LDH prior to ECMO cannulation and formed the study cohort (Fig. 1).

In this cohort, 122 patients (32%) had a high pre-ECMO LDH. Patients with a high LDH were more often male (76.2% vs 64.9%, $p = 0.026$), experienced lower rates of pre-ECMO prone positioning (72.1% vs 81.9%, $p = 0.045$), and had lower values of pH prior to ECMO cannulation (7.25 vs 7.33, $p < 0.001$) than those with low LDH. Similarly, patients with high LDH had higher platelet count (254 vs 222 k/uL, $p = 0.024$), serum creatinine (1.06 vs 0.84 mg/dL, $p < 0.001$), ferritin levels (1665 vs 935.6 ng/mL, $p < 0.001$), and lactate levels (2.2 vs 1.7 mmol/L, $p = 0.002$) (Table 1).

Hemorrhagic Stroke and ECMO

Forty-two (10.9%) of the patients suffered HS during ECMO. This translated into a rate of 0.36 events per 100 days of ECMO support. Patients who experienced HS did not show any significant differences in major demographics and cannulation strategies used, but had a lower pH (7.23 vs 7.32, $p = 0.002$), higher white blood cell count (WBC) (16.9 vs 14.0 k/uL, $p = 0.047$), higher serum creatinine before cannulation (1.24 vs 0.90 mg/dL, $p < 0.001$), and longer ECMO duration (33.0 vs 21.0 days, $p = 0.01$) than those without HS (Table 2). The overall on-support mortality was 38.6%, and HS was associated with higher mortality at the end of the follow-up period (82.9% vs 43.2%, $p < 0.001$). The most

common causes of mortality included multiorgan failure, stroke (ischemic or hemorrhagic) and cardiac failure.

Hemorrhagic Stroke and LDH

Twenty-one out of 122 (17%) patients with a high LDH experienced HS in comparison to 21 out of 262 (8%) with a low LDH ($p = 0.007$). Patients that experienced a HS, had a higher LDH prior to ECMO placement (737 vs. 589 U/L, $p = 0.021$, Fig. 2).

At 100 days after ECMO initiation, the probability of a HS was 40% in the high LDH group and 23% in those with a low LDH, $p = 0.002$ (Fig. 3A).

After adjustment for age, sex, prone positioning, ECMO duration, and pre-ECMO serum creatinine, patients in the high LDH group retained a greater likelihood of an eventual HS in comparison to those with low LDH (aHR, 2.40, 95% CI 1.27–4.51, $p = 0.007$). Findings remained consistent in separate multivariable models with adjustment for age, sex, pre-ECMO pH and lactic acid (aHR, 2.24, 95% CI 1.15–4.38, $p = 0.018$), age, sex, pre-ECMO platelets and ferritin (aHR, 3.30, 95% CI 1.65–6.59, $p = 0.001$, Fig. 3B). In subgroup analysis of patients cannulated for venovenous ECMO, those with a high LDH retained a greater risk of an eventual HS, (aHR 2.40, 95% CI [1.28, 4.47], $p = 0.007$), which remained similar in adjusted models (Fig. 4).

Discussion

We report a multicenter retrospective study examining the occurrence of HS and its relation to LDH during ECMO support for severe COVID-19 infection. The major findings of this study are as follows: 1) The occurrence of HS during ECMO for COVID-19 is elevated at 10.9%, 2) patients experiencing HS are at a substantially greater risk of death and 3) an elevated LDH over 750 U/L prior to ECMO placement is associated with an almost three-fold increase in developing HS during device support. This association between LDH and HS persisted after adjusting for confounders. Additionally, findings were preserved after limiting our analysis to patients cannulated for VV-ECMO.

HS remains a relatively common complication during ECMO with a remarkably strong association with mortality [9, 10, 19]. Recent studies show that the risk of HS can be substantially higher among COVID-19 patients compared to non-COVID-19 patients undergoing ECMO support [19, 30]. Seelinger et al. conducted a retrospective multicenter study among COVID-19 patients on ECMO, noting an HS incidence of 20%. Their study revealed a six-fold increased risk of HS compared to similar non-COVID-19 patients receiving ECMO for viral infection [19]. Our study showed a lower incidence of HS at 10.9%, in a relatively younger

Table 1 Baseline characteristics of adults supported with ECMO for COVID-19 stratified by LDH levels before ECMO cannulation

Baseline characteristics	Total <i>n</i> = 384	High LDH (> 750 U/L) <i>n</i> = 122	Low LDH (≤ 750 U/L) <i>n</i> = 262	<i>p</i> -value
Demographics				
Age, years—median [IQR]	48.0 [38.0–56.0]	46.0 [37.0–55.0]	49.0 [39.0–57.0]	0.32
Male sex, <i>n</i> (%)	263 (68.5%)	93 (76.2%)	170 (64.9%)	0.026
BMI, kg/m ² —median [IQR]	33.0 [29.3–38.9]	32.6 [30.1–38.5]	33.5 [29.0–39.2]	0.97
Race, <i>n</i> (%)				0.135
Caucasian	211 (56.1%)	58 (48.7%)	153 (59.5%)	
Black	65 (17.3%)	22 (18.5%)	43 (16.7%)	
Asian	18 (4.8%)	9 (7.6%)	9 (3.5%)	
Other	82 (21.8%)	30 (25.2%)	52 (20.2%)	
Hispanic ethnicity, <i>n</i> (%)	163 (44.7%)	43 (37.7%)	120 (47.8%)	0.07
Preexisting comorbidities, <i>n</i> (%)				
Hypertension	152 (64.7%)	39 (60.9%)	113 (66.1%)	0.46
Diabetes mellitus	118 (50.4%)	29 (46.0%)	89 (52.1%)	0.41
COPD	12 (5.1%)	1 (1.6%)	11 (6.4%)	0.13
Active smoking	18 (7.7%)	7 (10.7%)	11 (6.4%)	0.25
Organ transplant recipient	11 (4.7%)	2 (3.1%)	9 (5.3%)	0.49
CAD	13 (5.5%)	3 (4.7%)	10 (5.9%)	0.72
Pre-ECMO parameters				
Prone positioning, <i>n</i> (%)	302 (79.1%)	89 (72.1%)	213 (81.9%)	0.045
Cardiac Arrest, <i>n</i> (%)	56 (14.9%)	21 (17.4%)	35 (13.8%)	0.36
MAP, mmHg—median [IQR]	80.3 [71.6–90.0]	81.7 [71.8–90.5]	79.7 [71.7–89.3]	0.35
Vasopressors, <i>n</i> (%)	239 (63.2%)	74 (61.2%)	165 (64.2%)	0.56
P:F ratio—median [IQR]	74.0 [61.0–96.0]	76 [59.3–95.0]	73.5 [62.0–97.0]	0.74
pH—median [IQR]	7.31 [7.22–7.38]	7.25 [7.17–7.33]	7.33 [7.25–7.40]	<0.001
PaCO ₂ , mmHg—median [IQR]	57.0 [46.0–71.0]	61.5 [46.2–73.0]	56.0 [46.0–69.3]	0.10
WBC, k/uL—median [IQR]	13.2 [9.8–20.2]	14.7 [9.5–21.6]	14.0 [10.0–19.5]	0.86
Platelet Count, k/uL—median [IQR]	245 [182–331]	254 [191–332]	222 [166–311]	0.024
Creatinine, mg/dL—median [IQR]	0.91 [0.64–1.37]	1.06 [0.80–1.58]	0.84 [0.60–1.25]	<0.001
Bilirubin, mg/dL—median [IQR]	0.60 [0.40–0.80]	0.60 [0.50–0.90]	0.50 [0.30–0.80]	0.001
Ferritin, ng/mL—median [IQR]	1102 [570–1905]	1665 [931–2919]	935.6 [486–1563]	<0.001
CRP, mg/L—median [IQR]	22.6 [8.3–66.9]	27 [11.5–102.4]	19.3 [7.2–56.1]	0.043
INR—median [IQR]	1.20 [1.08–1.3]	1.2 [1.10–1.37]	1.18 [1.07–1.29]	0.018
Fibrinogen, mg/dL—median [IQR]	592.5 [417–700]	543 [322–678]	612 [477–715]	0.004
Lactic acid, mmol/L—median [IQR]	1.8 [1.3–2.6]	2.2 [1.46–3.3]	1.7 [1.2–2.4]	0.002
Use of systemic heparin on ECMO, <i>n</i> (%)	277 (73.7%)	92 (75.4%)	185 (72.8%)	0.59
Use of DTI on ECMO, <i>n</i> (%)	147 (39.2%)	49 (40.8%)	98 (38.4%)	0.65
Type of ECMO Used, <i>n</i> (%)				0.67
VV-ECMO	372 (96.9%)	118 (96.7%)	254 (97.0%)	
Cannulation strategy				0.66
Femoral vein-femoral vein	49 (13.2%)	20 (17.0%)	29 (11.5%)	
Femoral vein-jugular vein	171 (46.1%)	53 (44.9%)	118 (46.6%)	
Protek-Duo	75 (20.2%)	21 (17.8%)	54 (21.3%)	
Single right internal jugular vein	76 (20.5%)	24 (20.3%)	52 (20.6%)	
VA-ECMO	8 (2.1%)	2 (1.6%)	6 (2.3%)	
VAV-ECMO	4 (1.0%)	2 (1.6%)	2 (0.7%)	
ECMO Duration, days—median [IQR]	22.0 [11.0–42.0]	18.0 [10.0–34.0]	24.0 [13.0–43.0]	0.03
On-support mortality, <i>n</i> (%)	148 (38.6%)	49 (40.5%)	99 (37.8%)	0.61
Cause of death, <i>n</i> (%)				0.59
Cardiac failure	18 (12.2%)	8 (16.3%)	10 (10.2%)	

Table 1 (continued)

Baseline characteristics	Total <i>n</i> = 384	High LDH (> 750 U/L) <i>n</i> = 122	Low LDH (≤ 750 U/L) <i>n</i> = 262	<i>p</i> -value
Hemorrhage	12 (8.2%)	6 (12.2%)	6 (6.1%)	
Liver failure	1 (0.7%)	0 (0%)	1 (1.0%)	
Respiratory failure	17 (11.6%)	5 (10.2%)	12 (12.2%)	
Septic Shock	11 (7.5%)	2 (4.1%)	9 (9.2%)	
Stroke	21 (14.3%)	9 (18.4%)	12 (12.2%)	
Multi-organ failure	54 (36.7%)	16 (32.7%)	38 (38.8%)	
Mesenteric ischemia	3 (2.0%)	0 (0%)	3 (3.1%)	
Withdrawal of care	8 (5.4%)	2 (4.1%)	6 (6.1%)	
Anoxic brain injury	3 (2.0%)	1 (2.0%)	1 (1.0%)	
Intracranial hemorrhage, <i>n</i> (%)	42 (10.9%)	21 (17.2%)	21 (8.0%)	0.007

Comparisons made with Mann–Whitney *U* test for continuous variables and chi-squared for categorical variables

BMI body mass index, *COPD* chronic obstructive pulmonary disease, *CAD* coronary arterial disease, *DTI* direct thrombin inhibitors, *MAP* mean arterial pressure, *PaCO₂* arterial pressure of CO₂, *WBC* white blood cell, *CRP* C-reactive protein, *INR* international normalized ratio, *VV-ECMO* venovenous extracorporeal membrane oxygenation, *VA-ECMO* venoarterial extracorporeal membrane oxygenation, *VAV-ECMO* venoarterial-venous extracorporeal membrane oxygenation

Number of missing observations: WBC 1

Platelet: 6, Bilirubin 11, Ferritin 37, CRP 57, INR 33, Fibrinogen 146, Lactic acid 30, pH 3

population with less impairment of perfusion and hemodynamics at baseline [14, 19].

Numerous studies have elucidated risk factors for HS during ECMO in non-COVID-19 populations. Some of these factors include pre-ECMO hyperoxemia, higher inflammatory markers, thrombocytopenia, worse renal function, vasopressor use, higher lactate, and rapid decreases in PaCO₂ after ECMO initiation [10, 15, 17, 31–34]. We also noted that patients who experienced a HS, at baseline had worse renal function and acidosis. Reverberation of these risk factors with HS during refractory COVID-19 infection, indicates that a greater severity of underlying illness and inflammation are culprit in triggering HS. Importantly, we identify LDH as a readily available biomarker that may independently capture this inflammatory milieu and risk of eventual cerebral bleeding. The magnitude of LDH elevation in our cohort may reflect a more severe inflammatory state induced by COVID-19 infection, further demonstrated by the higher levels of other inflammatory markers, including ferritin and C-reactive protein.

LDH is an intracellular enzyme that gets released with cellular injury due to inflammatory conditions such as COVID-19 [26, 35]. Its use in prognostication has been established in many critical conditions including patients with COVID-19 [20, 25, 35–37]. However, evidence supporting its role in predicting HS remains uncertain. LDH levels can be elevated in cerebrospinal fluid of patients with stroke [38]. Its elevation is associated with worse outcomes in patients with both ischemic stroke and intracranial hemorrhage [24, 25, 39]. Chu et.al determined that LDH levels predict hematoma expansion in patients with HS using a

proteomic analysis [26], a finding that was later confirmed by Wang et al. in a much larger study of about 360 patients [27]. Despite these studies relating LDH to risk stratification for eventual neurologic injury during ECMO in patients with COVID-19 has not been established previously.

The mechanisms by which LDH may risk stratify patients for HS could be related to the underlying inflammatory response that accompanies severe COVID-19 cases requiring ECMO. The inflammatory cascade can result in progressive endothelial damage of brain vessels leading to the disruption of the blood–brain barrier [26, 27]. In COVID-19 specifically, it is known that the inflammatory response is exacerbated and is a major driver of the subsequent complications. Endotheliopathy involving intracranial vessels together with anticoagulation and ECMO-induced coagulopathy may constitute a microenvironment leading to an increased risk for HS [19, 40–42]. Our findings indicate that LDH can serve as a prognostic marker prior to ECMO cannulation, useful for risk stratification, that could help clinicians' decision-making for optimal ECMO candidacy and expectations before cannulation. Additional prospective studies are needed to validate our findings and determine whether patients higher pre-ECMO LDH would incur a lower burden of HS with minimization of anticoagulation during ECMO during COVID-19 and beyond.

Limitations

Our study has a several limitations. Given the low number of HS in our cohort, the multivariable analysis could not be adjusted for all covariates at once to prevent overfitting

Table 2 Baseline characteristics of adults supported with ECMO for COVID-19 stratified by occurrence of hemorrhagic stroke

Baseline characteristics	Total <i>n</i> = 384	Hemorrhagic Stroke <i>n</i> = 42	No Hemorrhagic Stroke <i>n</i> = 342	<i>p</i> -value
Demographics				
Age, years—median [IQR]	48 [38.0–56.0]	50.5 [43.0–58.0]	47.0 [37.0–56.0]	0.27
Male sex, <i>n</i> (%)	263 (68.5%)	27 (64.3%)	236 (69.0%)	0.53
BMI, kg/m ² —median [IQR]	33 [29.3–38.9]	33.6 [30.6–39.0]	33.0 [29.2–38.8]	0.49
Race, <i>n</i> (%)				0.16
Caucasian	211 (56.1%)	19 (47.5%)	192 (57.1%)	
Black	65 (17.3%)	10 (25.0%)	55 (16.4%)	
Asian	18 (4.8%)	4 (10.0%)	14 (4.2%)	
Other	82 (21.8%)	7 (17.5%)	75 (22.3%)	
Hispanic ethnicity, <i>n</i> (%)	163 (44.7%)	26 (63.4%)	176 (54.3%)	0.27
Comorbidities, <i>n</i> (%)				
Hypertension	152 (64.7%)	17 (73.9%)	135 (63.7%)	0.32
Diabetes mellitus	118 (50.4%)	10 (45.5%)	108 (50.9%)	0.62
COPD	12 (5.1%)	0 (0.0%)	12 (5.6%)	0.24
Active smoking	18 (7.7%)	1 (4.4%)	17 (8.1%)	0.52
Organ transplant recipient	11 (4.7%)	1 (4.4%)	10 (4.7%)	0.93
CAD	13 (5.5%)	1 (4.4%)	12 (5.7%)	0.79
Pre-ECMO parameters				
Prone Positioning, <i>n</i> (%)	382 (79.1%)	30 (71.1%)	272 (80.0%)	0.20
Cardiac Arrest, <i>n</i> (%)	56 (14.9%)	6 (15.0%)	50 (14.9%)	0.99
MAP, mmHg—median [IQR]	80.3 [71.6–90.0]	81.3 [73.7–90.3]	80.3 [71.7–89.8]	0.89
Vasopressor Use, <i>n</i> (%)	239 (63.2%)	30 (71.4%)	209 (62.2%)	0.24
P:F ratio—median [IQR]	74.0 [61.0–96.0]	80.2 [66.0–97.0]	73.0 [60.0–96.0]	0.24
pH—median [IQR]	7.31 [7.22–7.38]	7.23 [7.15–7.31]	7.32 [7.23–7.39]	0.002
PaCO ₂ , mmHg—median [IQR]	57.0 [46.0–71.0]	60.0 [47.0–73.0]	57.0 [45.0–70.0]	0.27
WBC, k/uL—median [IQR]	13.2 [9.8–20.2]	16.9 [12.2–24.9]	14.0 [9.8–19.2]	0.047
Platelet Count, k/uL—median [IQR]	245.0 [182.0–331.0]	231.0 [153.0–346.0]	248.0 [183.5–325.5]	0.57
Creatinine, mg/dL—median [IQR]	0.91 [0.64–1.37]	1.24 [0.83–1.69]	0.90 [0.63–1.30]	<0.001
Bilirubin, mg/dL—median [IQR]	0.60 [0.40–0.80]	0.50 [0.40–1.00]	0.60 [0.40–0.80]	0.72
Ferritin, ng/mL—median [IQR]	1102 [570–1905]	1121.0 [648.0–1672.0]	1102.0 [563.5–1933.0]	0.83
CRP, mg/L—median [IQR]	22.6 [8.3–66.9]	30.2 [8.3–102.4]	21.9 [8.4–61.1]	0.72
INR—median [IQR]	1.20 [1.08–1.3]	1.10 [1.07–1.20]	1.20 [1.09–1.30]	0.27
Fibrinogen, mg/dL—median [IQR]	592.5 [417–700]	576.0 [435.0–696.0]	594.0 [413.0–700.0]	0.90
Lactic acid, mmol/L—median [IQR]	1.80 [1.30–2.60]	1.75 [1.30–2.30]	1.80 [1.30–2.60]	0.53
Use of Systemic Heparin on ECMO, <i>n</i> (%)	277 (73.7%)	30 (73.2%)	247 (73.7%)	0.93
Use of DTI on ECMO, <i>n</i> (%)	147 (39.2%)	17 (40.5%)	130 (39.0%)	0.85
Type of ECMO Used, <i>n</i> (%)				0.77
VV-ECMO	372 (96.9%)	41 (97.6%)	331 (96.8%)	0.29
Cannulation strategy				
Femoral vein-femoral vein	40 (13.2%)	5 (12.2%)	44 (13.3%)	
Femoral vein-jugular vein	171 (46.1%)	14 (34.2%)	157 (47.6%)	
Protek-Duo	75 (20.2%)	13 (31.7%)	62 (18.8%)	
Single right internal jugular vein	76 (20.5%)	9 (22.0%)	67 (20.3%)	
VA-ECMO	8 (2.1%)	1 (2.4%)	7 (2.0%)	
VAV-ECMO	4 (1.0%)	0 (0.0%)	4 (1.2%)	
ECMO Duration, days—median [IQR]	22.0 [11.0–42.0]	33.0 [19.0–55.0]	21.0 [10.0–40.0]	0.01

Comparisons made with Mann–Whitney U test for continuous variables and chi-squared for categorical variables

BMI body mass index, *COPD* chronic obstructive pulmonary disease, *CAD* coronary arterial disease, *DTI* direct thrombin inhibitors, *MAP* mean arterial pressure, *PaCO₂* arterial pressure of CO₂, *WBC* white blood cell, *CRP* C-reactive protein, *INR* international normalized ratio, *VV-ECMO* venovenous extracorporeal membrane oxygenation, *VA-ECMO* venoarterial extracorporeal membrane oxygenation, *VAV-ECMO* venoarterial-venous extracorporeal membrane oxygenation

Number of missing observations: WBC 1

Platelet: 6, Bilirubin 11, Ferritin 37, CRP 57, INR 33, Fibrinogen 146, Lactic acid 30, pH 3

Fig. 2 A comparison of pre-ECMO lactate dehydrogenase level between patients with and without an eventual hemorrhagic stroke

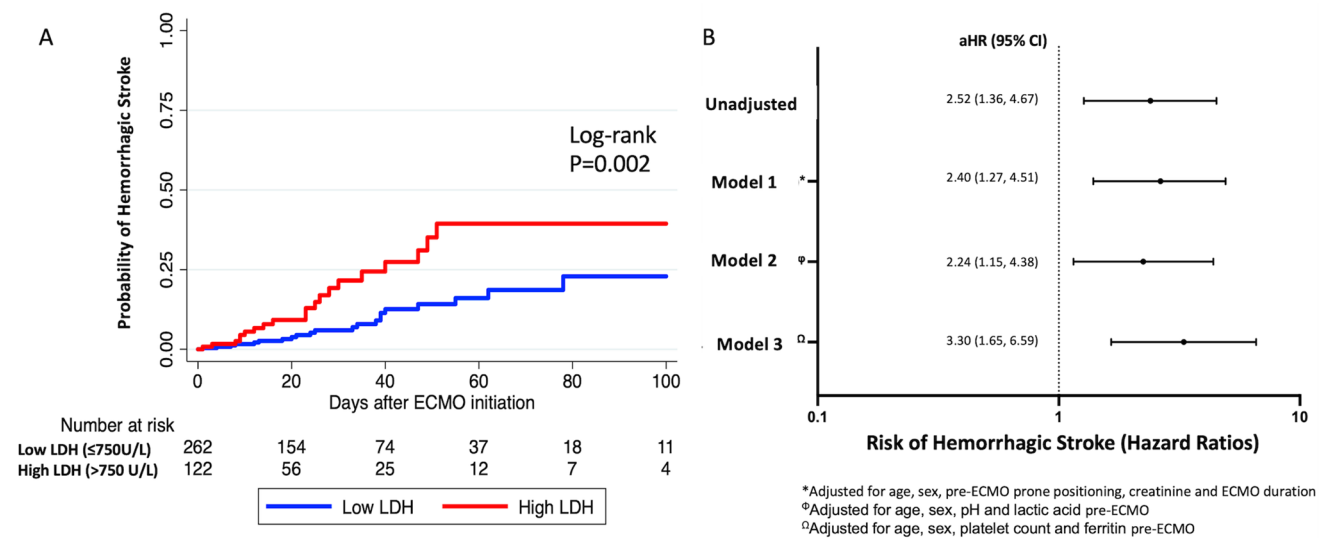
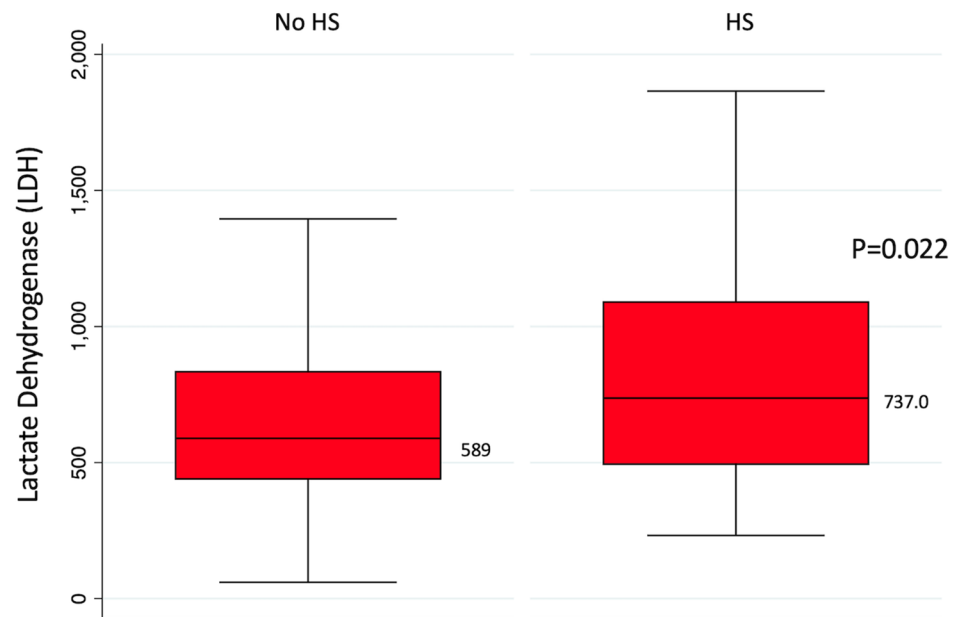


Fig. 3 **A** Kaplan Meier curves comparing likelihood of a hemorrhagic stroke in patients with COVID-19 supported by ECMO stratified by a high or low pre-ECMO lactate dehydrogenase. ECMO,

extracorporeal membrane oxygenation. **B.** Unadjusted and adjusted hazard ratios of a hemorrhagic stroke in patients with a pre-ECMO lactate dehydrogenase > 750 U/L

the model. Unmeasured confounders not included in the adjustment model could have introduced bias, but these would be present in a non-differential manner in LDH groups. Unfortunately, data on ECMO pump equipment or coagulation parameters at the time of the event were not

available in the dataset. Moreover, data on LDH changes after ECMO cannulation leading up to occurrence of HS were not captured as part of the data collection. Lastly, the retrospective nature of our study makes it improbable to derive causation.

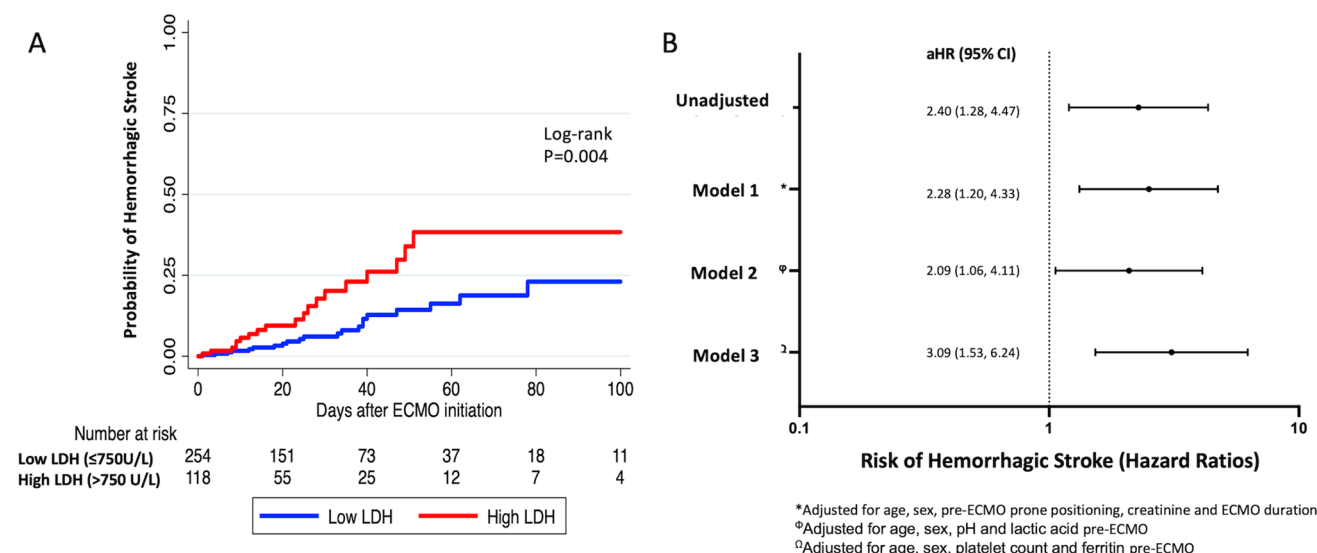


Fig. 4 **A** Kaplan Meier curves comparing likelihood of a hemorrhagic stroke in patients with COVID-19 supported by venovenous ECMO stratified by a high or low pre-ECMO lactate dehydrogenase.

ECMO, extracorporeal membrane oxygenation. **B.** Unadjusted and adjusted hazard ratios of a hemorrhagic stroke in patients with a pre-ECMO lactate dehydrogenase > 750 U/L

Conclusion

An elevated LDH > 750 U/L prior to ECMO initiation is associated with a higher risk of HS during device support in patients with COVID-19. Further prospective studies are warranted to confirm these findings and to examine whether reduction in anticoagulation in these patients can reduce the burden of brain hemorrhage.

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Declarations

Competing interests The authors declare no competing interests.

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